EchoST-SAM: Semantic-Prompted Spatiotemporal LV Segmentation for Functional Analysis in Echocardiography

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November 19, 2026

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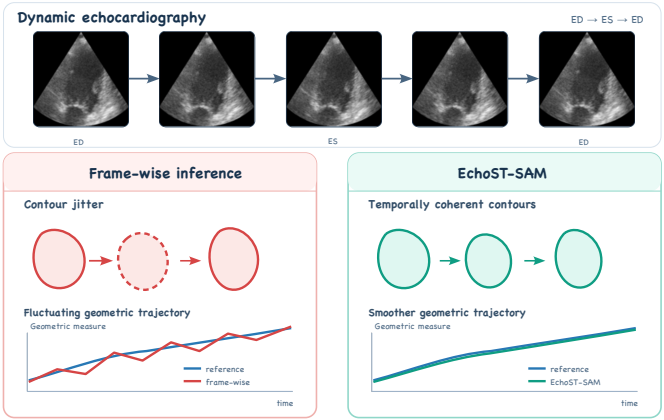
Overview

1. Introduction and Motivation
2. EchoST-SAM Method
3. Experiments and Results
4. Conclusions

Research question

Can automatic anatomical prompting and temporal modeling improve LV contours and the geometry derived from them?

LV segmentation across a cardiac cycle



The trajectories are conceptual illustrations.

- Speckle and weak boundaries make individual frames ambiguous.
- Independent predictions can produce **contour jitter**.
- Contour errors change the measured cavity shape and area.

Goal: accurate contours with coherent evolution over time.

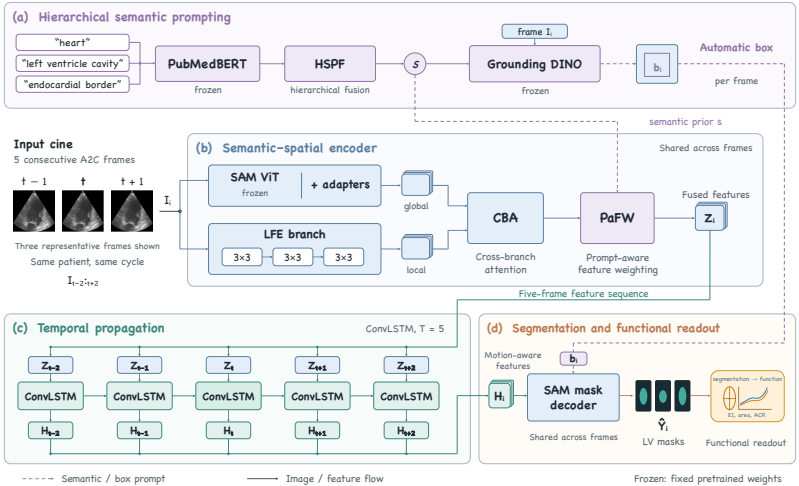
Related work and our approach

Research direction	Representative approaches
Frame-level models	U-Net, TransUNet, Swin-Unet
SAM adaptation	MedSAM, SAMUS, CC-SAM
Temporal models	Temporal correction, MediViSTA

EchoST-SAM

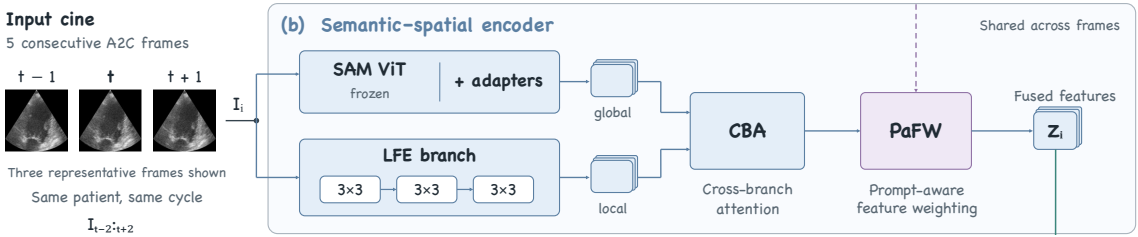
- Automatic anatomical prompting with global–local features.
- ConvLSTM propagation before SAM mask decoding.
- Joint evaluation of segmentation, consistency and geometry.

EchoST-SAM architecture



Anatomical guidance, spatial encoding and temporal propagation share one segmentation pipeline.

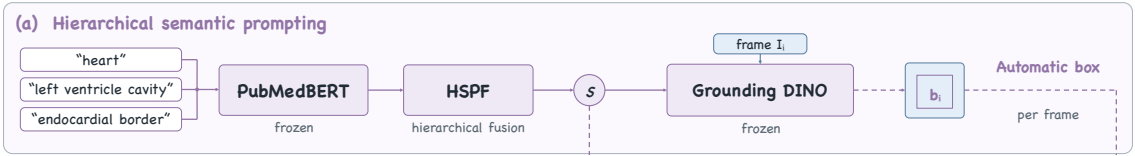
Global and local visual features



- **Global branch:** frozen SAM ViT with trainable feature and position adapters.
- **Local branch:** standard and dilated 3×3 convolutions.
- **Cross-branch attention:** global queries attend to local keys and values.

Both branches produce 32×32 feature maps from 256×256 frames.

Anatomical guidance without manual prompts



A shared semantic prior

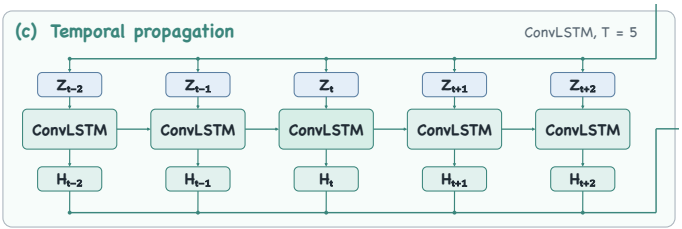
- Frozen PubMedBERT encodes the three fixed phrases.
- HSPF averages and projects their embeddings.
- The cached prior is the same across images.

Two uses of anatomical guidance

- Grounding DINO generates a **frame-specific box**.
- PaFW weights spatial tokens by cosine alignment with the prior.

No user-provided clicks or boxes during segmentation.

Temporal propagation and supervision



$$(H_t, C_t) = \mathrm{ConvLSTM}(Z_t, H_{t-1}, C_{t-1}), \qquad \hat{Y}_t = M_{\mathrm{dec}}(H_t, s_t)$$

- Five consecutive frames per clip. Reset both states at each clip.
- Preserve frame order and share the SAM decoder across frames.
- Supervise every frame with BCE, Dice loss and PaFW entropy regularization.

Geometry derived from the masks

Ellipse eccentricity (EI)

$$\mathrm{EI}_t = \sqrt{1 - \left(\frac{L_{t,\min}}{L_{t,\max}}\right)^2}$$

Fit an ellipse to the largest contour.

EI describes **elongation** through the minor and major axes.

Fractional cavity-area change (ACR)

$$\mathrm{ACR} = \frac{A_{\mathrm{ED}} - A_{\mathrm{ES}}}{A_{\mathrm{ED}}}$$

Count foreground pixels to obtain area.

ACR describes **relative area reduction** in the A2C view.

Scope of the reported evaluation

Reference ED/ES indices isolate segmentation-derived error. These geometric surrogates do not replace biplane LVEF or clinical assessment.

Dataset and evaluation protocol

CAMUS A2C

- 500 patients, apical two-chamber recordings.
- Official ED/ES masks plus clinician annotations for intermediate frames.
- Study-specific patient-disjoint split.

Train	Validation	Test
400	50	50

Complementary endpoints

- Dice and maximum Hausdorff distance.
- Temporal consistency (TC): Dice between adjacent predicted masks.
- EI and ACR mean absolute errors.
- Correlation and Bland–Altman agreement.

All methods share the split, dense labels and 256×256 resolution. Validation selects checkpoints and window length.

Lower geometric errors with a small Dice trade-off

Method	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
U-Net	93.56	9.90	0.0167	0.082
TransUNet	93.60	9.60	0.0155	0.080
AAU-Net	93.59	9.13	0.0160	0.078
CC-SAM	92.85	10.88	0.0150	0.074
EchoST-SAM (spatial)	**93.97**	**9.05**	0.0142	0.071
EchoST-SAM	93.90	9.11	**0.0131**	**0.064**

Full model versus the spatial variant

EI MAE: **7.7% lower** ACR MAE: **9.9% lower**
Dice: **0.07 percentage points lower**

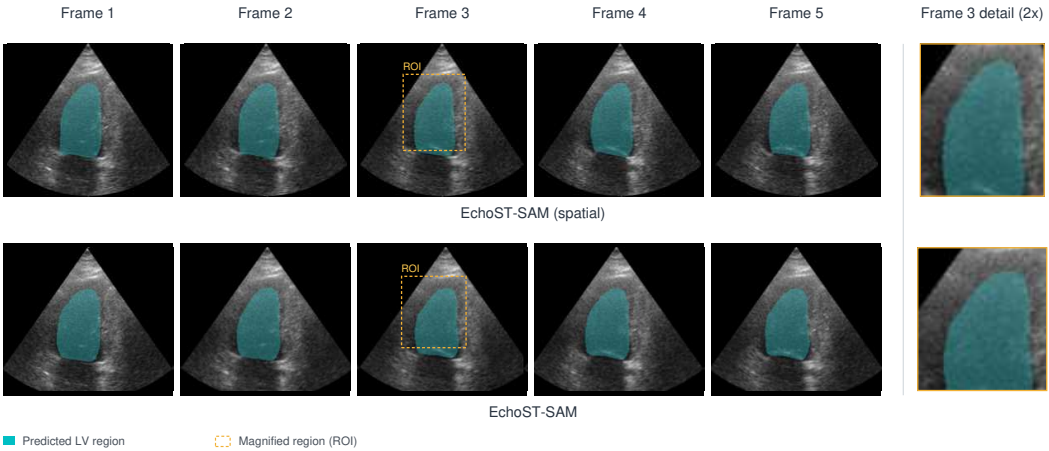
Selected external baselines shown. Full comparison in the backup slides. Single-run point estimates.

Five-frame propagation improves adjacent-mask consistency

Window T	Dice (%) ↑	HD (px) ↓	TC ↑
1 frame (spatial)	**93.97**	**9.05**	0.921
3 frames	93.84	9.28	0.944
5 frames (default)	93.90	9.11	0.956
7 frames	93.76	9.33	**0.957**

- $T = 1$ to $T = 5$: **TC increases by 0.035**.
- $T = 7$ adds only 0.001 TC and lowers frame-level accuracy.
- TC can reward over-smoothing. Interpret it alongside Dice and geometric errors.

Contour evolution over five consecutive frames



Smoother contour evolution in this example. Cyan marks predictions. The rightmost column magnifies frame 3.

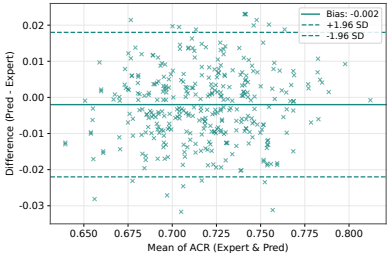
Spatial components contribute to segmentation accuracy

Spatial variant	Dice (%) ↑	HD (px) ↓
Without local feature enhancement	92.60	10.02
Without semantic prompting	92.84	9.88
Without HSPF	93.08	9.63
Without prompt-aware weighting	93.21	9.52
Full spatial model	**93.97**	**9.05**

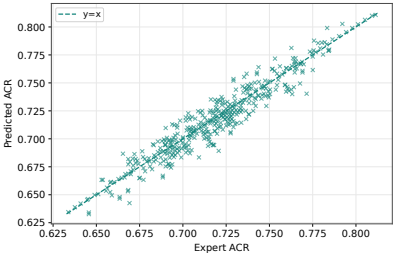
- ConvLSTM is disabled for every row.
- Removing LFE gives the largest Dice decrease: **1.37 percentage points**.
- A matched non-semantic prior is needed to isolate the role of biomedical meaning.

Agreement of mask-derived geometric measurements

	Pearson r		Mean bias	
	Spatial	Full	Spatial	Full
EI	0.924	**0.937**	$+0.003$	$+0.002$
ACR	0.931	**0.945**	-0.005	-0.002



(c) Bland-Altman Plot for ACR



(d) Correlation Plot for ACR (r=0.945)

ACR plots for the full model. Small mean bias does not establish clinical interchangeability.

Limitations and next steps

- **Generalization**
  One CAMUS A2C setting. Test additional views, centers and acquisition conditions.
- **Uncertainty**
  Single-run point estimates. Quantify uncertainty at the patient level.
- **Temporal validity**
  Add temporal baselines and ground-truth consistency. Assess over-smoothing and error propagation.
- **Functional scope**
  Validate automatic ED/ES selection and clinically relevant endpoints beyond EI and ACR.

Conclusions

- EchoST-SAM combines anatomical guidance, adapted SAM features and temporal propagation for interaction-free LV segmentation.
- CAMUS A2C: **93.90% Dice**, **9.11 px HD**, **0.956 TC**.
- Temporal modeling lowers EI/ACR errors to **0.0131/0.064**, with a **0.07-percentage-point decrease in Dice**.

Thank you!

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Project page: `https://lvyupku.github.io/EchoST-SAM/`

Full comparison on CAMUS A2C

Method	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
U-Net	93.56	9.90	0.0167	0.082
Swin-Unet	91.88	13.11	0.0192	0.087
MISSFormer	93.25	9.94	0.0163	0.084
TransUNet	93.60	9.60	0.0155	0.080
AAU-Net	93.59	9.13	0.0160	0.078
H2Former	93.44	9.60	0.0153	0.077
MedSAM	87.52	15.28	0.0239	0.091
SAMed	87.67	13.24	0.0242	0.094
SAMUS	91.13	11.76	0.0172	0.079
CC-SAM	92.85	10.88	0.0150	0.074
EchoST-SAM (spatial)	**93.97**	**9.05**	0.0142	0.071
EchoST-SAM	93.90	9.11	**0.0131**	**0.064**

Single-run point estimates. The spatial variant excludes ConvLSTM.

Full spatial component ablations

Variant	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
Without LFE	92.60	10.02	0.0167	0.083
Without semantic prompting	92.84	9.88	0.0162	0.081
Without HSPF	93.08	9.63	0.0154	0.079
Without PaFW	93.21	9.52	0.0150	0.078
Full spatial model	**93.97**	**9.05**	**0.0142**	**0.071**

- ConvLSTM is disabled in every configuration.
- LFE: local feature enhancement.
- HSPF: pooling and projection of the fixed anatomical text embeddings.
- PaFW: prompt-aware feature weighting.

Training configuration and trainable components

Optimization

- 200 epochs, batch size 16.
- Adam, learning rate 10^{-2}.
- Weight decay 10^{-4}.
- StepLR: multiply by 0.1 every 50 epochs.
- One NVIDIA RTX 4090.

Trainable

Adapters, LFE, HSPF, PaFW, ConvLSTM and mask decoder.

Frozen

SAM ViT backbone, PubMedBERT, Grounding DINO and SAM prompt encoder.

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T}\left[\mathcal{L}_{\mathrm{BCE}}(\hat{Y}_t, Y_t) + \mathcal{L}_{\mathrm{Dice}}(\hat{Y}_t, Y_t) + 0.1\,\mathcal{H}(w_t)\right]$$

$$\mathcal{H}(w_t) = -\sum_{j} w_{t,j} \log w_{t,j}$$

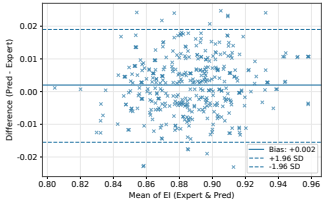
Minimizing entropy encourages concentrated token weights.

Metric definitions and evaluation scope

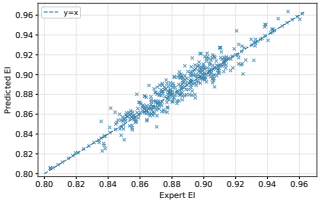
$$\mathrm{TC} = \frac{1}{T_c - 1} \sum_{t=1}^{T_c - 1} \mathrm{Dice}(\hat{Y}_t, \hat{Y}_{t+1})$$

- TC averages adjacent predicted-mask overlap within each cine, then across test cines. T_c is cine length; $T = 5$ is the clip window.
- HD is the **symmetric maximum** Hausdorff distance in pixels, not HD95.
- EI measures ellipse elongation, not the clinical short-axis eccentricity ratio.
- ACR uses **reference ED/ES indices** for the reported evaluation.
- EI/ACR compare prediction-derived and annotation-derived geometry. Frames from the same patient are dependent.

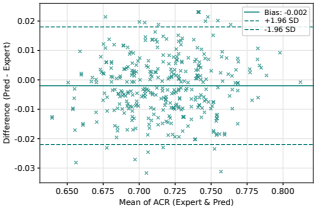
Full EI and ACR agreement plots



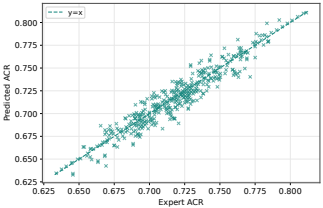
(a) Bland-Altman Plot for EI



(b) Correlation Plot for EI (r=0.937)



(c) Bland-Altman Plot for ACR



(d) Correlation Plot for ACR (r=0.945)

Full EchoST-SAM model; predicted versus expert-derived geometry.

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Acknowledgments

- National Natural Science Foundation of China
  Grant Nos. 62576222 and 62266039
- Department of Education of Guangdong Province
  Grant No. 2023ZDZX2055
- Shenzhen Medical Research Fund
  Grant No. D250402003
- Shenzhen Science and Technology Program
  Grant Nos. KJZD20240903095605007 and JCYJ20250604145033042
- China Scholarship Council (CSC)

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